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MILITARY SPECIFICATION

IMPREGNITE, UNSTABILIZED (CC2)

AND STABILIZED (XXCC3)

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-I-285B. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers two types of impregnite.

1.2 Classification. Impregnite shall be of the following types as specified (see 6.2):

Type I = Unstabilized: N,N'-dichloro-bis (2,4,6-trichlorophenyl) urea (CC2), made by coupling trichloroaniline hydrochloride with urea to form sym-bis (2,4,6-trichlorophenyl) urea with subsequent chlorination.

Type II - Stabilized: type I impregnite with zinc oxide, micronized (XXCC3).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.

FSC 6850

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MILITARY

MIL-Z-291	-	Zinc Oxide for Chemical Warfare Use (Pyrotechnics or Stabilizer).
MIL-D-14258	-	Drum, Fiber, 20-Gallon Capacity, for Chemicals.
MIL-P-15011	-	Pallets, Material Handling, Hardwood, Post Construction, 4-Way.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads, 40" x 48", 4-Way (Partial) Pallet Skids, Runners or Pallet-type Base.

DRAWINGS

US ARMY MUNITIONS COMMAND

C6-16-1 - Drum, Fiber, 20-Gallon - Assembly, Detail and Bill of Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-79 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are a part of the Code of Federal Regulations (revised 1966) available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 71-79 (Rev. 1966).")

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distrubuted among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Particle size characteristics.

3.1.1 Type I. No type I impregnite shall be retained on a 2000-micron (No. 10) sieve when tested as specified in 4.3.4.1(a).

3.1.2 Type II. Type II impregnite shall conform to the particle size characteristics of table I when tested as specified therein.

Table I. Particle size characteristics

Particle size	Minimum percent by wt	Test paragraph
Less than 44 microns (passing through a No. 325 sieve)	99	4.3.4.1(a)
Less than 8 microns	80	4.3.4.1(b)
Less than 6 microns	65	4.3.4.1(b)
Less than 4 microns	45	4.3.4.1(b)

3.2 Chemical characteristics.

3.2.1 Type I. Type I impregnite shall be prepared from sym-bis (2,4,6-trichlorophenyl) urea which is free of other chlorinatable or oxidizable material and shall conform to the chemical characteristics of table II when tested as specified therein.

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Table II. Chemical characteristics of type I impregnite

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Volatile matter	----	0.20	4.3.4.2
Chloroform insoluble (dry basis)	----	4.25	4.3.4.3
Chlorides as NaCl (dry basis)	----	0.50	4.3.4.4
Active chlorine (dry, chloroform-soluble basis)	14.2	14.6	4.3.4.5

3.2.2 Type II. Type II impregnite shall be a mixture of the type I impregnite specified in 3.2.1 and zinc oxide conforming to grade B, class 1 of MIL-Z-291 and shall conform to the chemical characteristics of table III when tested as specified therein. No other additives shall be used.

Table III. Chemical characteristics of type II impregnite

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Volatile matter	----	0.18	4.3.4.2
Active chlorine (dry, chloroform-soluble basis)	12.8	13.4	4.3.4.5
Zinc oxide content	8.1	10.1	4.3.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Preproduction.

4.2.1 Sample. A preproduction sample of no less than 2000 pounds (lb) for type I impregnite and no less than 2100 lb for type II impregnite shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production. Three 1-lb representative specimens shall be randomly selected from the preproduction sample. Two of the three specimens shall be thoroughly mixed to form a composite specimen.

4.2.2 Inspection. Both the composite specimen and the remaining individual specimen shall be tested as specified in the applicable portions of 4.3.4. Failure of any test shall be cause for rejection of the preproduction sample.

4.3 Regular production.

4.3.1 Lotting. A lot of type I impregnite shall consist of that material produced by one manufacturer, at one plant, and under essentially the same manufacturing conditions in no more than 24 consecutive hours. A lot of type II impregnite shall consist of that material produced by one manufacturer, at one plant, and under essentially the same manufacturing conditions in one work shift.

4.3.2 Sampling.

4.3.2.1 For examination. Sampling for examination shall be conducted on the basis of containers and in accordance with MIL-STD-105.

4.3.2.2 For test.

(a) Lot. Four specimens of no less than one lb each shall be randomly selected from the lot. In the event of a continuous operation, a 1-lb specimen shall be taken at each 2-hour interval. All of the specimens except one shall be thoroughly mixed to form a composite specimen. The composite specimen and the remaining individual specimen shall be placed in separate clean, dry containers labeled to identify the lot from which they were taken.

(b) Batch. In the event of a batch operation (see 6.3), sampling shall be conducted on the basis of containers and in accordance with MIL-STD-105, level S-2 except that a minimum of three samples shall be taken. A specimen of no less than one lb shall be removed from each container in the sample. Approximately equal portions of all the specimens except one shall be thoroughly mixed to form a composite specimen. The composite specimen and the remaining individual specimen shall be placed in separate clean, dry containers labeled to identify the batch and containers from which they were taken.

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4.3.3 Inspection.

4.3.3.1 For examination. Sample containers and the level B preparation for delivery thereof shall be examined in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 For test. The composite specimen and the remaining individual specimen shall be tested in duplicate as specified in the applicable portions of 4.3.4. Failure of any test shall be cause for rejection.

4.3.3.3 Classification of defects.

(a) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Container not as specified
102	Container damaged or leaking
103	Container closure not as specified
104	Quantity per container not as specified
105	Marking of container incorrect, illegible, missing, or incomplete

4.3.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.3.4.1 Particle size.

(a) Sieve test. Nest an appropriate size sieve as specified in 3.1 and conforming to RR-S-366 with a receiving pan. Weigh 100.0 ± 0.05 gram (g) of the specimen, place on the sieve, and cover. Place the sieve, receiving pan, and cover in a single-eccentric type mechanical shaker geared to produce a rotary motion and tapping action at a uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Shake for 3 minutes, weigh the material on the receiving pan and calculate the percent retained or passing through the sieve as applicable.

(b) Pipet method. Determine the particle size of the impregnite in accordance with the "short" method in the appendix unless otherwise specified (see 6.2).

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4.3.4.2 Volatile matter.. Break up all lumps due to caking in the container, pass the entire specimen through a 590-micron (No.. 30) sieve conforming to RR-S-366, and return to the specimen container. Weigh to the nearest milligram (mg) approximately 5 g of the specimen in a tared, glass-stoppered weighing bottle approximately 50 millimeters (mm) high. Remove the stopper and dry for 4 hours in an oven at 50° to 55°C and under a partial vacuum corresponding to no more than 8 mm of mercury. Cool in a desiccator and weigh. Calculate the percent volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100(A-B)}{A}$$

where: A = Weight of specimen before heating in grams and
B = Weight of specimen after heating in grams.

4.3.4.3 Chloroform insoluble.

(a) Solvent. Use U.S.P. or reagent grade chloroform and purify it on the same day it is to be used as follows: Place 1 liter of the chloroform in a 2-liter separatory funnel and shake with four changes of water, 500 milliliters (ml) each. Filter the washed chloroform through two thicknesses of dry filter paper. Use chloroform which does not give a blank when 50 ml are tested as described in 4.3.4.5. If a blank greater than 0.20 ml of 0.1N sodium thiosulfate is obtained, repurify the chloroform as above, adding 10 g of potassium iodide in the first wash water. In case of dispute, retest using reagent grade chloroform purified on the same day it is used as follows: Place 1 liter of the chloroform in a 2-liter separatory funnel and shake with three changes of water, 500 ml each. Filter the washed chloroform through dry filter paper and dry for 1 hour over calcium chloride with occasional shaking. Remove the chloroform from the calcium chloride and distill, discarding the first 100 ml of distillate.

(b) Procedure. Weigh to the nearest milligram approximately 2 g of the sample and transfer to a 250-ml beaker, add 100 ml of the purified chloroform prepared as specified in (a), and stir thoroughly. Filter through a dry, carefully weighed Gooch crucible, using suction, and wash the beaker and residue with an additional 100 ml of the purified chloroform. Dry to constant weight at 105° to 110°C, cool in a desiccator, and weigh. Retain the crucible and residue for the determination of chlorides (4.3.4.4). Calculate the percent chloroform insoluble as follows:

$$\text{Percent chloroform insoluble (dry basis)} = \frac{(10000)A}{(100-B)C}$$

where: A = Weight of insoluble matter in grams,
B = Percent volatile matter (see 4.3.4.2), and
C = Weight of specimen in grams.

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4.3.4.4 Chlorides.

(a) Potassium thiocyanate solution. Dissolve 10 g of potassium thiocyanate in water and dilute to 1 liter. The solution is adjusted by titration against 0.02N silver nitrate solution. It is advisable to have 1 ml of the thiocyanate equivalent to 1 ml of the silver nitrate solution. Owing to the deliquescence of the thiocyanates, the exact amount for a 0.02N solution cannot be weighed.

(b) Ferric ammonium sulfate indicator. Dissolve 80 g of clear crystals of ferric ammonium sulfate in water and make up to 1 liter. A few drops of sulfuric acid may be added if necessary to clear up the solution.

(c) Procedure. Wash the residue obtained in the determination of chloroform insoluble (see 4.3.4.3), together with the asbestos mat from the crucible, into a 300-ml flask with 100 ml of water. Shake the flask vigorously for 3 to 5 minutes to break up the mat and dissolve the chlorides. Add 2 to 3 ml of concentrated nitric acid, mix, and determine the chloride by the Volhard method. Calculate the percent chlorides as NaCl (dry basis) as follows:

$$\text{Percent chlorides} = \frac{584(AB-CD)}{E(100-F)}$$

where: A = Milliliters of silver nitrate solution used to titrate specimen,
B = Normality of silver nitrate solution,
C = Milliliters of potassium thiocyanate solution used,
D = Normality of potassium thiocyanate solution,
E = Weight of specimen in grams, and
F = Percent volatile matter (see 4.3.4.2).

4.3.4.5 Active chlorine. Weigh from 0.550 to 0.600 g of the sample and wash through a funnel into a 300-ml glass-stoppered Erlenmeyer flask with 50 ml of the purified chloroform (see 4.3.4.3). Stopper the flask and agitate gently to dissolve the specimen. As soon as the specimen has dissolved, add a freshly prepared potassium iodide solution consisting of 4 ml of water, 2 g of potassium iodide, and 5 ml of 1 to 1 hydrochloric acid (dissolve the potassium iodide in the water, then add the hydrochloric acid). Stopper the flask and shake vigorously for 5 minutes. Cool the flask in cold water, open, and wash the stopper and walls of the flask with water. Titrate the liberated iodine with 0.1N sodium thiosulfate solution. During the titration, swirl the contents of the flask continuously and vigorously so there will always be an excess of iodine in the acid-aqueous layer until the final end point is reached. Near the end point, stopper and shake the flask vigorously after each addition of thiosulfate. When the iodine color has almost disappeared, add starch indicator and titrate to the disappearance of the blue color. Run a blank on 50 ml of the chloroform by the same procedure. Calculate the percent active chlorine (on dry, chloroform-solution basis) as follows:

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$$\text{Percent active chlorine} = \frac{17730(A-B)C}{W(100-D)(100-E)}$$

where: A = Milliliters of sodium thiosulfate solution used to titrate the specimen,
B = Milliliters of sodium thiosulfate solution used to titrate the blank,
C = Normality of sodium thiosulfate solution,
D = Percent volatile matter (see 4.3.4.2),
E = Percent chloroform insoluble (see 4.3.4.3), and
W = Weight of specimen in grams.

4.3.4.6 Zinc oxide content (type II impregnite).

(a) Standard solution. Dissolve 32.31 g of potassium ferrocyanide and 0.3 g of potassium ferricyanide in water and make up to a volume of 1 liter. Standardize the solution by titration against a 0.3 g sample of pure metallic zinc, or equivalent quantity of pure zinc oxide, in accordance with the method specified in (b). Calculate the weight of zinc oxide equivalent to 1.00 ml of the solution.

(b) Determination. Weigh to the nearest milligram approximately 1 g of type II impregnite and transfer the specimen to a 400-ml beaker. Add 15 ml of methyl alcohol, 6 ml of concentrated hydrochloric acid, and 25 ml of water. Place the beaker on a steam bath and heat for 15 minutes. At the end of the 15-minute heating period, add 200 ml of hot water, transfer the beaker to a hot plate, and with constant stirring, add 3 ml of cencentrated ammonium hydroxide and 2 drops of a 0.25 percent solution of ferrous ammonium sulfate. Titrate the hot (75° to 95°C) solution with the standard potassium ferrocyanide - potassium ferricyanide solution until the color changes from blue to greenish white. Run a blank determination and calculate percent zinc oxide as follows:

$$\text{Percent zinc oxide} = \frac{100\ (A-B)C}{W}$$

where: A = Milliliters of standard solution used for specimen,
B = Milliliters of standard solution used for blank,
C = Grams of zinc oxide equivalent to 1 ml of standard solution, and
W = Weight of specimen in grams.

5. PREPARATION FOR DELIVERY

5.1 Packing, level B.

5.1.1 Type I impregnite. Seventy-five lb ± 1/2 lb of type I impregnite shall be packed in a fiber drum conforming to MIL-D-14258. The drum shall be closed and sealed as shown on Drawing C6-16-1.

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5.1.2 Type II impregnite. Seventy-five lb ± 1/2 lb of type II impregnite shall be packed, closed, and sealed in a fiber drum as specified for the type I impregnite in 5.1.1.

5.2 Marking. In addition to the precautionary marking specified in 5.2.1 and any special marking required by the contract or order, marking for drums shall be in accordance with Interstate Commerce Commission regulations and MIL-STD-129 (see 6.2).

5.2.1 Precautionary marking. Each drum shall be marked as shown on Drawing C6-16-1 and as follows:

(a) Using 2-inch high letters, place the marking "DO NOT STORE ABOVE 120°F" in three locations on the drum: approximately 4 inches above the regular marking, diametrically opposite, and on the top of the drum.

(b) Using letters of 1-inch minimum height, mark the drum as follows: "KEEP AWAY FROM OPEN FIRES AND ALL HEATED SURFACES; AVOID INHALATION OF FUMES."

(c) The precautionary markings specified in (a) and (b) shall not interfere or obscure other markings on the drum.

5.3 Palletization. When specified in the contract or order, drums shall be palletized in accordance with the requirements for load type XIII of MIL-STD-147 utilizing pallets conforming to MIL-P-15011.

6. NOTES

6.1 Intended use. Type I and type II impregnite are intended for use in impregnating clothing used to protect personnel against the action of vesicant-type chemical agents.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of impregnite required.
(c) Special marking other than 5.2.1, if required.
(d) Palletization, if required.
(e) If a method other than the "short" method in the appendix is to be used for determining particle size (micron range) of type II impregnite.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant places in Specified Limiting Values (ASTM E29).

6.5 Precautions.

6.5.1 Handling. Mixtures of finely ground impregnite and zinc oxide are flammable and all necessary precautions should be taken for protection of personnel.

6.5.2 Storage. Prolonged storage of impregnite without proper ventilation is conducive to spontaneous combustion. Thermal decomposition of type II impregnite is easily initiated and if it is packed in thin walled containers and compactly stored in a closed room, the heat transferred from one container to the next is sufficient to propagate the decomposition, thereby causing a storage hazard. In order to minimize storage hazards the following precautions should be observed:

(a) Railroad cars of type II impregnite should not be allowed to stand in the general vicinity of the micronizing plant.

(b) Filled drums of type II impregnite should be placed on a pallet with the drums positioned so that they will be no less than six inches apart and no less than six inches from a wall.

(c) If loaded pallets of type II impregnite are stacked in tiers, the vertical space between drums should be alternately 6 inches and 24 inches, with 6 inches between drums in the first and second tiers. Supports should be provided for the pallets so that no tier of containers will support more than one tier of containers stacked above it.

(d) Place a placard carrying a warning to keep the type II impregnite away from open fire and all heated surfaces on or near each exposed side of storage area.

(e) Pallets should be in accordance with 5.3.

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APPENDIX

DETERMINATION OF PARTICLE SIZE OF TYPE II IMPREGNITE

(PIPET METHOD)

10. SCOPE

10.1 This appendix describes a method for determining the particle size of micronized type II impregnite using a sedimentation pipet.

20. APPLICABLE DOCUMENTS

SPECIFICATIONS

MILITARY

MIL-P-256 - Polyvinyl Alcohol, Granular.

30. PREPARATION OF SOLUTION

30.1 Preparation of a 3-1/3-percent polyvinyl alcohol (PVA) solution.

30.1.1 Ingredients. The ingredients required for preparation of sufficient 3-1/3-percent PVA solution are as follows: 2900 g of water, 1 g of emulsifying agent ("Duponol G" or equivalent), and 100 g of polyvinyl alcohol conforming to grade B, class 1 of MIL-P-265.

30.1.2 Procedure. Dissolve 1 g of "Duponol G" in 900 g of water in a 2-liter beaker with a minimum of stirring. Slowly add 100 g of PVA with agitation. In order to avoid lump formation, the PVA should be slowly sprinkled in over a period of about 30 minutes. After 30 minutes of additional stirring, place the beaker on a steam bath and heat to 65° to 70°C with continued stirring. Shut the steam off and continue stirring for about an hour. Dilute the resulting solution with the remainder of the water (2000 g) in a large stoppered bottle to make the 3-1/3-percent PVA solution. After thorough mixing, allow the solution to stand for several hours. Only the relatively clear supernatant liquid should be used; any undissolved residue at the bottom should be discarded.

30.2 Preparation of 1-percent dispersion of type II impregnite. Prepare a 1-percent dispersion of type II impregnite as follows:

(a) Weigh 7.50 g of dry type II impregnite into a 50-ml beaker.

(b) Fill a buret with 3-1/3-percent PVA.

(c) Run 5.0 ml of 3-1/3-percent PVA into the 50-ml beaker.

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(d) Stir contents with a glass rod until "breaking" occurs, i.e., the mixture takes on first a dough-like and then a syrupy consistency.

(e) Continue stirring for an additional 5 minutes, taking care to work in any dry material adhering to the sides of the beaker and rod without, however, exerting any appreciable grinding action.

(f) During stirring, rotate both the beaker and rod in the hands.

(g) Add 5.0 ml 3-1/3-percent PVA and stir for 1 minute.

(h) Add 12.5 ml of 3-1/3-percent PVA and stir for 1 minute (a total of 22.5 ml of PVA solution is used).

(i) Rinse contents of small beaker into a large 32-ounce (oz) stoppered bottle using 720 ml of water. Do not scrape the dry material off the edge of beaker.

(j) Finally "rock the bottle" by hand for 3 minutes (the "rocking" procedure is described in section 50).

40. CALIBRATION OF SEDIMENTATION BOTTLES

40.1 The sedimentation bottle, shown on figure 1, is doubly calibrated so as to give the weight of suspension which will fill the assembled apparatus to two different heights above the pipet tip, the distance being 20.00 centimeters (cm) and 5.00 cm, respectively. Two initial calibrations are made by determining the weights of water that fill the assembled bottle to the above-mentioned levels. In each case, the calibration is checked by adding to the sedimentation bottle a weight of a 1-percent suspension of type II impregnite which is 30 g greater than the initial calibration value obtained with water. Three 10-ml samples are then withdrawn by means of the pipet, using the procedure given in section 60 for withdrawal of samples by means of the aspirator. About 10 minutes later, the foam will have subsided sufficiently so that the true level of the liquid can be determined. If the resulting level is not exactly right (i.e., at 20.00 cm or 5.00 cm), the requisite weight of the 1-percent type II impregnite dispersion is corrected to make it so. If the sedimentation bottle is constructed according to the scale specified, about 250 g of 1-percent type II impregnite suspension will be required for the 5.00 cm level and about 650 g for the 20.00 cm level. The 20.00 cm level is used in the "long method" for characterization of the complete particle size distribution, whereas the 5.0 cm level is used for the "short method" for cutting at 8, 6, and 4 microns. When the sedimentation bottle is once correctly calibrated for 1-percent type II impregnite slurry, subsequent filling operations are simplified.

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50. FILLING SEDIMENTATION BOTTLE WITH 1-PERCENT TYPE II IMPREGNITE SUSPENSION

50.1 The 1-percent suspension of type II impregnite contained in the 32-oz stoppered bottle is always again homogenized just prior to filling the sedimentation container by "rocking the bottle" as follows: invert the bottle about 135° from its normal vertical position and shake until all sediment on the bottom end disappears. With the bottle held in a horizontal position, continue the mixing by "rocking" (i.e., slowly lifting alternate ends of the bottle about 45° from the horizontal every 2 seconds) about a dozen times. A quantity of the homogenized slurry as determined by the above-described calibration is weighed into a tared beaker from whence it is immediately poured into the sedimentation bottle upon which the pipet is assembled, the bottle then being clamped in a vertical position.

60. DETAILED PROCEDURE FOR WITHDRAWING A SAMPLE

60.1 The detailed order of procedure is as follows, the letters referring to the parts of the pipet indicated on figure 1. Particular attention should be given to the necessity for withdrawing the samples at a uniform rate and by a constant procedure.

(a) Turn stopcock (connecting B and C) to drain pipet.

(b) See that leveling bulb of aspirator is on upper ring.

(c) Connect rubber tube of aspirator to top end of pipet.

(d) Raise leveling bulb momentarily to expel drainage through (C).

(e) Close stopcock.

(f) Lower leveling bulb to lower ring support which is clamped at a level which will require about 25 seconds for filling the pipet (see section 100).

(g) Open stopcock (connecting B and A) and draw sample into pipet. As the sample level approaches the 10.00 ml calibration line, the stopcock is gradually turned off so that flow can be stopped exactly at the mark. (If the level is inadvertently passed, the situation can be remedied by temporarily raising the leveling bulb of the aspirator and allowing the liquid level to come back to the proper mark slowly.)

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(h) With the stopcock turned off, the aspirator leveling bulb is raised to the upper ring.

(i) Place coded sample container under the outlet (C).

(j) Open stopcock (connecting B and C) and allow contents of pipet to drain into sample container.

(k) If necessary to expel the last few drops of sample, a momentary lifting of the leveling bulb above the upper support ring will accomplish the result, the inside of the sample container being touched to the outlet (C).

(l) Close the stopcock.

(m) Lower the leveling bulb to the lower ring.

(n) Place a small beaker of rinse water so that (C) is below the surface of the water.

(o) Again open the stopcock (connecting B and C).

(p) Allow the rinse water to be sucked into the pipet bulb until the latter is about 2/3 filled.

(q) Remove the beaker at this point and allow air to bubble through the pipet for a few seconds (being careful that liquid is not sucked into the aspirator rubber tube).

(r) Close stopcock.

(s) Place aspirator leveling bulb on the upper ring support.

(t) Replace sample container under (C).

(u) Open stopcock (connecting B and C).

(v) Allow rinse water to drain into sample bottle, raising leveling bulb momentarily if necessary.

(w) Close stopcock.

(x) Remove sample container for analysis of active chlorine content.

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70. PROCEDURE FOR WITHDRAWING THE THREE CONTROL SAMPLES (FOR EITHER THE "LONG" OR "SHORT" METHOD).

70.1 With the 1-percent suspension of type II impregnite in the assembled sedimentation bottle, proceed as follows:

(a) "Rock the bottle" by holding the assembled pipet in both hands with the left hand around the large ground glass joint, thumb over the breather hold (J) and with the right hand over the bottom end of the bottle. The actual "rocking" action to be applied to the pipet bottle is the same as described above for the 32-ounce stoppered bottle.

(b) Clamp the pipet bottle in a vertical position and take the first 10.00 ml control sample after about a minute, using the pipet and aspirator as directed in section 60.

(c) Code the sample container "Test Number - 1."

(d) Again "rock the bottle" and take the second control sample as before, labeling the sample container "Test Number - 2."

(e) "Rock the bottle" a third time and take the last control sample, labeling the sample container "Test Number - 3."

80. PROCEDURE FOR MAKING THE COMPLETE TEST

80.1 Having decided whether the "short" method (i.e., 3 cuts at 5.00 cm) or the "long" method (i.e., 9 cuts at 20.00 cm) is to be used, proceed as follows:

(a) Fill sedimentation bottle with homogenized slurry to calibrated level determined as explained in section 70.

(b) Take three successive 10-ml control samples of homogenized slurry for analysis as explained in section 70.

(c) Have sample containers coded according to the particular schedule selected (see sample data sheets that follow).

(d) Set clock at "a very local time" of 11:58.

(e) Homogenize the slurry by "rocking the bottle."

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(f) Set the sedimentation bottle down at 12:00 (zero time) and clamp firmly in a vertical position in a place free from drafts, radiation, and excessive vibration.

(g) Open stopcock to let liquid seek its level in the capillary, being certain that no rinse drains into the bottle.

(h) Note temperature.

(i) Take samples at times indicated by the code on the sample containers using the same technique as that described in section 70 for taking the three control samples. Care is exercised in this case, however, that the slurry is not disturbed other than by the simple action of drawing up the 10-ml sample. Between sampling, the suspension in the capillary is not allowed to seek its level. The rinse water drainage should be blown out of the outlet cock before each sample is drawn up.

(j) Titrate each sample for active chlorine by the method described in section 90.

(k) Taking the average of the three titers of the control samples as representing 100 percent of the type II impregnite occupying 10 ml of homogenized slurry, calculate the "percent undersize" of the subsequent sample cuts by comparing the respective titers of the samples with respect to the average of the control titers.

(l) Record the data obtained on the data sheet.

(m) If the sedimentation was not carried out at 25°C, correct the results to this temperature by increasing the apparent micron diameter to the extent of 1.3 percent for each 1° under 25°C and by decreasing the apparent micron diameter to the extent of 1.0 percent for each degree over 25°C.

(n) The results are conveniently expressed by plotting on graph paper of the ordinary type. However, a plot on "Logarithmic Probability" paper is preferred (with "micron size" on the logarithmic scale and "percent undersize" and "percent oversize" on the "probability" scale). The resulting curve approaches a straight line and occasional errors may be more readily disclosed.

(o) The "percent undersize" material at any given particle size can be interpolated from the graph.

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90. TITRATION OF ACTIVE CHLORINE CONTENT OF SAMPLES

90.1 Take the sample in the 300 ml Erlenmeyer flask and add 30 ml of purified chloroform and 20 ml of glacial acetic acid. Insert the stopper and agitate gently for a moment to disperse the sample. Add 45 ml of 15-percent potassium iodide solution, stopper the flask, place a little water in the lip of the flask around the stopper to seal the ground joint, and shake vigorously for about 10 seconds. Place the flask in a beaker of cold water for several minutes to reduce the pressure in the flask so that the stopper will not blow out. Remove the flask from the beaker, fill the lip with water, and carefully remove the stopper. Wash down the stopper and the sides of the flask with water and titrate the sample with 0.1N standard sodium thiosulfate solution. Insert the stopper and shake the flask vigorously by hand at frequent intervals during the titration. When the stopper is removed after each shaking, wash it and the sides of the flask with water. Do not add the thiosulfate in a quantity large enough to competely discharge the brown color of iodine from the water layer. Vigorous shaking will transfer the iodine from the water layer to the solvent layer. When the purple color is almost discharged from the solvent layer and the water layer has a slight light straw color, add 10 ml of starch solution and continue the titration until the purple color is changed to white by the addition of one drop of thiosulfate. As the endpoint is approached, shake the flask vigorously after the addition of each drop of thiosulfate. Run a blank determination on the reagents and correct the sample titer for the blank obtained.

100. NOTE

100.1 More complete details on the pipet method for the determination of particle size will be found in "Pipette Method for Determining the Particle Size of CC2", March 10, 1943, N.D.R.C. Misc. Pub. 283.

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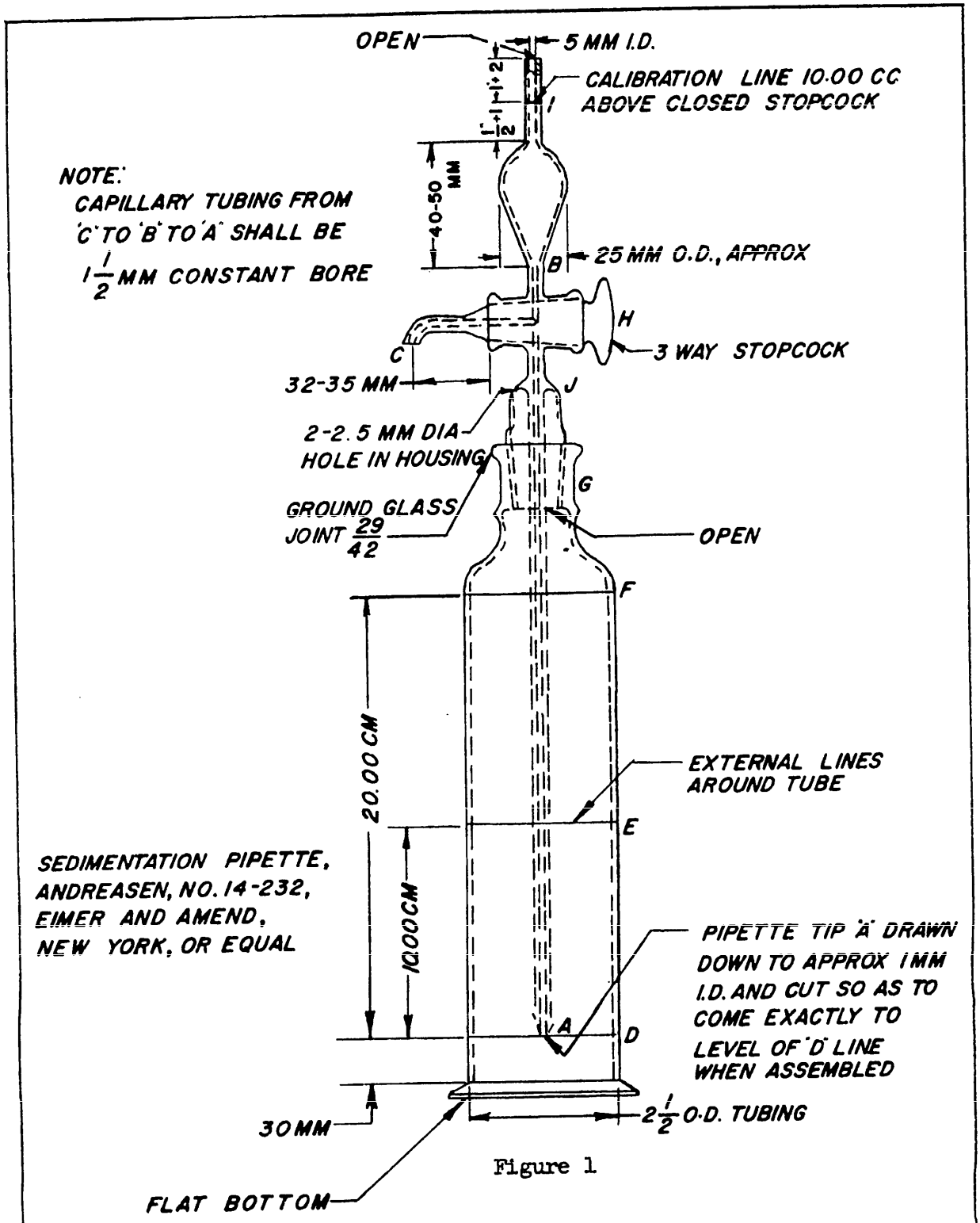


Figure 1

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Sample data sheet for "short method" with 5.00 cm height

Test No. X

Date started _____

Sample _____

Pipet No. _____ Normality of $Na_2S_2O_3$ _____

Weight of 1-percent slurry _____ grams Analyst _____

Apparent micron	Temp. (°C) and	Sample code				Percent	Micron diameter corrected
		Test					
Diameter	Remarks	No.	Hours	Minutes	Titer	Under-size	to 25°C
∞		X	1		---	100	∞
∞		X	2	Controls	---	100	∞
∞		X	3		---	100	∞
8		X	0	29	---	---	---
6		X	0	47.5	---	---	---
4		X	1	38	---	---	---

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Sample data sheet for "long method" with 20.00 cm height

Test No. X

Date started _____

Sample _____

Pipet No. _____ Normality of $Na_2S_2O_3$ _____

Weight of 1-percent slurry _____ grams Analyst _____

Apparent micron	Temp. (°C) and	Sample code				Percent	Micron diameter corrected
		Test					
Diameter	Remarks	No.	Hours	Minutes	Titer	Under-size	to 25°C
∞		Y	1		---	100	∞
∞		Y	2	Controls	---	100	∞
∞		Y	3		---	100	∞
15.0		Y	0	34	---	---	---
12.5		Y	0	48	---	---	---
10.0		Y	1	13	---	---	---
8.75		Y	1	34	---	---	---
7.50		Y	2	06	---	---	---
6.25		Y	2	57	---	---	---
5.00		Y	4	31	---	---	---
4.00		Y	6	54	---	---	---
2.16		Y	23	20	---	---	---

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